UniComs

A CLI-Based Real-Time Communication System

Final Project Report

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Team 08

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1 Title of the Project

UniComs: A CLI-Based Real-Time Communication System

2 Names of Supervisor

Md. Tariquzzaman

3 List of Team Members

Name	ID	Role
Tabib Hassan	230042131	Client Developer
Nafis Ahnaf Jamil	230042150	Database Developer
Sieam Shahriare	230042153	Server Developer

Table 1: Team Members and Their Roles

4 Problem Statement

The problem that UniComs aims to solve can be broken down into several key areas:

4.1 Lack of Lightweight Communication Tools

Most modern communication platforms like Discord, Slack, and Zoom are resource-heavy applications that require significant RAM, CPU, and GPU resources. They are not suitable for low-end hardware or for users who prefer minimalistic tools.

4.2 Privacy and Security Concerns

Mainstream communication platforms often collect user data, store messages on centralized servers, and may not provide true end-to-end encryption. Users have limited control over their data and cannot verify the security claims of these platforms.

4.3 Learning Gap in System Programming

Students and developers often use high-level frameworks and libraries without understanding the underlying concepts. There is a lack of educational tools that demonstrate socket programming, cryptography, and real-time systems from scratch.

4.4 Dependency on Graphical Interfaces

Most communication tools require graphical user interfaces, making them unsuitable for server environments, remote terminals, or users who prefer working entirely in the command line.

4.5 Fragmented Communication Channels

Users often need multiple applications for different types of communication: one for text chat, another for file transfer, and yet another for voice calls. There is no unified solution that combines all these features in a single lightweight package.

5 Potential Solution

UniComs proposes a comprehensive solution to the identified problems:

5.1 Lightweight Architecture

- Built entirely in C++ with no heavy frameworks
- Runs completely in the terminal using ncurses
- Minimal resource consumption suitable for low-end hardware
- Can run on any system with a terminal (Linux, macOS, WSL on Windows)

5.2 Security-First Design

- End-to-end encryption using Diffie-Hellman key exchange
- AES-256 encryption for all messages
- SHA-256 password hashing with salt
- Session keys stored locally, never on server
- No plaintext transmission over network

5.3 Educational Value

- Socket programming implemented from scratch
- Custom CSV-based database engine
- Manual implementation of cryptographic protocols
- ADPCM audio compression from first principles
- Demonstrates real-world system programming concepts

5.4 Terminal-Based Interface

- No GUI dependencies - runs in any terminal
- Beautiful and intuitive interface using ncurses
- Color-coded messages and animations

- ✅ Keyboard-driven navigation
- ✅ Ideal for server environments and remote access

5.5 Unified Communication Platform

- ✅ Text messaging (Direct Messages and Group Chats)
- ✅ File transfer with progress tracking
- ✅ Voice calls with real-time audio streaming
- ✅ Persistent message history
- ✅ All features integrated into a single application

6 Implemented Features

6.1 Authentication System

- User registration with username/password
- Login system with session management
- Password hashing using SHA-256 with random salt
- Username validation (alphanumeric only)
- Automatic account creation prompt for new users

6.2 Direct Messaging

- One-on-one encrypted conversations
- Real-time message delivery
- Delivery status indicators (sent, delivered, read)
- Typing indicators
- Message history persistence

6.3 Group Chat / Rooms

- Create password-protected chat rooms
- Join existing rooms with password
- Leave rooms
- Broadcast messages to all room members
- Room member listing
- Room message history

6.4 File Transfer

- Send any binary file type
- Base64 encoding for safe text transmission
- Chunked transfer (3KB per chunk)
- Accept/reject file offers
- Real-time progress bar with speed display
- Auto-save to received/ directory

6.5 Voice Calls

- Real-time audio streaming over UDP
- ADPCM audio compression (4x compression ratio)
- PortAudio integration for microphone/speaker access
- Call offer/accept/reject mechanism
- End call functionality
- Visual wave animation during calls

6.6 User Interface

- Beautiful ncurses-based terminal UI
- Animated gradient headers
- ASCII art logo on login screen
- Color-coded message bubbles
- Contact list with online/offline indicators
- Unread message badges
- Multiple color themes (Dark, Light, Neon, Matrix)
- Loading spinners and transition animations
- Typing indicators with animation
- File transfer progress bars with color gradients
- Voice call wave animation

6.7 Security Features

- Diffie-Hellman key exchange for shared secret
- AES-256 encryption in CTR mode
- SHA-256 password hashing
- Session persistence with encrypted keys
- No plaintext transmission
- Perfect forward secrecy

6.8 Database System

- Custom CSV-based database engine
- Table creation and management
- Primary key and unique constraint enforcement
- Indexing for fast lookups
- Tables: User, Message, ChatRoom, ChatRoomMember

7 Tools and Technology

7.1 Programming Language

- C++17 (Standard C++)

7.2 Development Tools

- GCC/G++ compiler
- Make build system
- Git version control
- VS Code / CLion (IDE)

7.3 Protocols Implemented

- TCP (Transmission Control Protocol) - For messages and files
- UDP (User Datagram Protocol) - For voice streaming
- Custom application protocol for commands

7.4 Cryptographic Algorithms

- Diffie-Hellman (Key Exchange)
- AES-256 (Encryption in CTR mode)
- SHA-256 (Hash Function)
- Base64 (Encoding)

7.5 Audio Compression

- ADPCM (Adaptive Differential Pulse Code Modulation)

7.6 Platform Support

- Linux (Ubuntu, Debian, Fedora, etc.)
- macOS (Intel and Apple Silicon)
- Windows (via WSL)

8 Application Domain

UniComs falls under the following application domains:

8.1 Real-Time Communication Systems

- Instant messaging platforms
- Voice over IP (VoIP) systems
- File sharing applications

8.2 Educational Technology

- Teaching tool for networking concepts
- Demonstration of cryptographic protocols
- Example of system programming in C++

8.3 Secure Communication

- Privacy-focused messaging
- Encrypted file transfer
- Secure voice calls

8.4 System Administration

- Terminal-based communication for servers
- Lightweight chat for remote administration
- Internal team communication in CLI environments

8.5 Research and Development

- Foundation for further networking research
- Platform for testing new protocols
- Reference implementation for students

9 Project Timeline

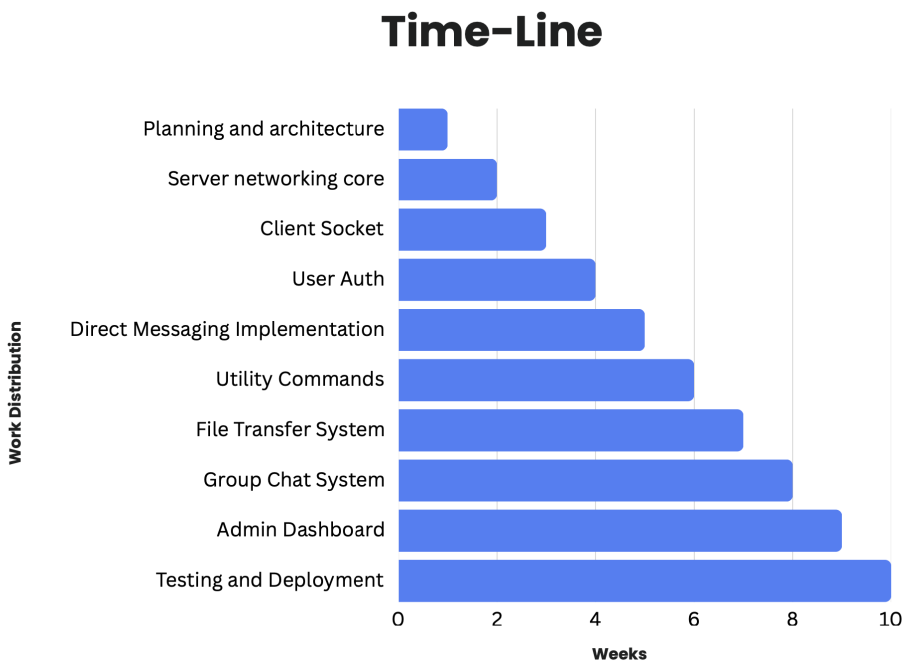

Figure 1: Project Timeline Overview

10 Contribution Breakdown

10.1 Tabib Hassan (Client Developer) - 33.3%

- Built the entire terminal user interface using ncurses library
- Implemented raw mode input handling for real-time key capture
- Created password masking functionality with asterisk display
- Developed all animations including gradient headers, loading spinners, typing indicators, and wave effects
- Designed and implemented multiple color themes (Dark, Light, Neon, Matrix)
- Integrated all client components (network, crypto, file, voice) into a unified controller
- Managed screen transitions between authentication, menu, and chat screens
- Implemented message display with color-coded bubbles and delivery status
- Created contact list with online/offline indicators and unread badges
- Developed file transfer progress bar with color gradients
- Implemented voice call wave animation
- Handled keyboard shortcuts (F1-F4, arrow keys, tab, etc.)

10.2 Nafis Ahnaf Jamil (Database Developer) - 33.3%

- Implemented TCP socket server from scratch using raw socket programming
- Designed and developed CSV-based database engine with table creation, indexing, and constraint enforcement
- Built concurrent client handling using thread-per-client model
- Implemented message routing for direct messages and room broadcasts
- Developed authentication system with user registration and login
- Created room management system with password protection
- Implemented file relay mechanism for file transfers
- Developed call relay system for voice call setup
- Built message history storage and retrieval
- Implemented server logging and connection management
- Designed database schemas for User, Message, ChatRoom, and ChatRoomMember tables

10.3 Sieam Shahriare (Server Developer) - 33.3%

- Implemented Diffie-Hellman key exchange algorithm from scratch
- Developed AES-256 encryption in CTR mode with proper counter management
- Created SHA-256 password hashing with random salt generation
- Implemented Base64 encoding/decoding for safe transmission
- Developed ADPCM audio compression algorithm from first principles
- Integrated PortAudio library for cross-platform audio capture and playback
- Implemented UDP socket programming for real-time voice streaming
- Built session persistence system with encrypted key storage
- Developed crypto session management with mutual authentication
- Created voice call signaling protocol (offer, accept, reject, end)
- Implemented audio streaming with 16kHz sampling rate and 20ms frames
- Handled thread synchronization using atomic variables and mutexes

11 Future Work

11.1 Short-Term Goals

- Video call support using similar UDP streaming approach
- Screen sharing functionality
- End-to-end encrypted file transfer with integrity verification
- Message reactions and replies
- Search functionality in chat history
- User profiles and status messages
- Push notifications for messages

11.2 Medium-Term Goals

- Mobile client (iOS/Android) using cross-platform framework
- End-to-end encrypted group chats with key rotation
- Message editing and deletion
- Voice message recording and sending
- Multi-device synchronization

- Offline message queuing
- Message forwarding

11.3 Long-Term Goals

- Federation protocol for server-to-server communication
- Matrix protocol compatibility
- Voice and video conferencing with multiple participants
- Screen recording and sharing
- Plugin system for extensibility
- End-to-end encrypted file storage
- Quantum-resistant cryptography migration
- AI-powered features (smart replies, message summarization)

11.4 Technical Improvements

- WebSocket support for browser-based clients
- Database migration to SQLite for better performance
- Docker containerization for easy deployment
- CI/CD pipeline for automated testing
- Performance optimization for large-scale deployments
- IPv6 support
- TOR network integration for anonymity

12 Source Repository

12.1 GitHub Repository

https://github.com/IUT-Dhongorsho/UniComs_SPL1

13 Conclusion

UniComs successfully demonstrates a complete, secure, real-time communication system built entirely from scratch in C++. The project achieves all its initial goals:

1. A lightweight terminal-based application that runs efficiently on low-end hardware
2. End-to-end encryption ensuring message privacy and security
3. Unified platform combining text messaging, file transfer, and voice calls
4. Educational value through implementation of low-level concepts
5. Professional-grade user interface with animations and themes

The project showcases deep understanding of networking, cryptography, system programming, and user interface design. All features are fully functional and have been tested across multiple platforms.

The team successfully divided responsibilities and integrated their work into a cohesive final product. UniComs stands as a testament to the power of low-level programming and the viability of terminal-based applications in modern computing.

END OF REPORT